# Introducing the circle

**ANSWER KEY**

## Core Concepts: Introducing the circle

**1** For the model $f(x) = 4x + 4$ in introducing the circle, evaluate $f(2)$.

**a.** $12$ ✓

**b.** $16$

**c.** $8$

**d.** $16$

Substitute $x = 2$: $f(2) = 4(2) + 4 = 12$.

**2** Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

**a.** $x = 1$ and $x = 4$

**b.** $x = 1$ **and** $x = 3$ ✓

**c.** $x = -1$ and $x = -3$

**d.** $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

**3** A data set recorded for introducing the circle consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** $16.5$

**b.** $17.5$

**c.** $18.5$ ✓

**d.** $20.5$

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by $4$ gives $18.5$.

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 4x + 4$ in introducing the circle, evaluate $f(3)$.

**a.** $16$ ✓

**b.** $20$

**c.** $12$

**d.** $20$

Substitute $x = 3$: $f(3) = 4(3) + 4 = 16$.

**5** Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ **and** $x = 4$ ✓

**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

**6** A data set recorded for introducing the circle consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** $16.5$

**b.** $17.5$

**c.** $18.5$ ✓

**d.** $20.5$

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by $4$ gives $18.5$.

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## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 4x + 4$ in introducing the circle, evaluate $f(4)$.

**a.** $20$ ✓

**b.** $24$

**c.** $16$

**d.** $24$

Substitute $x = 4$: $f(4) = 4(4) + 4 = 20$.

**8** Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

**a.** $x = 3$ and $x = 6$

**b.** $x = 3$ **and** $x = 5$ ✓

**c.** $x = -3$ and $x = -5$

**d.** $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

**9** A data set recorded for introducing the circle consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** $16.5$

**b.** $17.5$

**c.** $18.5$ ✓

**d.** $20.5$

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by $4$ gives $18.5$.

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## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 4x + 4$ in introducing the circle, evaluate $f(5)$.

**a.** $24$ ✓

**b.** $28$

**c.** $20$

**d.** $28$

Substitute $x = 5$: $f(5) = 4(5) + 4 = 24$.

**11** Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

**a.** $x = 4$ and $x = 7$

**b.** $x = 4$ **and** $x = 6$ ✓

**c.** $x = -4$ and $x = -6$

**d.** $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

**12** A data set recorded for introducing the circle consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** $16.5$

**b.** $17.5$

**c.** $18.5$ ✓

**d.** $20.5$

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by $4$ gives $18.5$.

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